



Department of Information Technology

Academic Year 2025 - 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Conversion of NFA into DFA

- **Course Outcome:** CO1

- **Topic Learning Outcome:** TLO3

- **Activity Chosen:** Collaborative Learning

- **Justification:**

In unit I, The **Conversion of NFA into DFA** is an important topic. The question asked in most of the university questions. A collaborative learning activity was chosen to promote active participation and peer learning and also clear the doubts. This activity enhanced teamwork, communication skills, and problem-solving abilities while allowing slow learners to benefit from peer support. The collaborative approach also helped identify misconceptions early and provided immediate feedback through group discussions.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- The topic **Conversion of NFA into DFA** was taught in the previous classes. The time allotted for the activity is 10 Minutes.
- A team of six members each team assigned with one problem statement as given in the topics such as **Conversion of RE to ϵ -NFA, ϵ -NFA to NFA, NFA into DFA** as shown in Figure 2.
- The team members discussed and completed the conversion as shown in Figure 1.
- Sample answer for the solutions is as shown in Figure 3 and 4. The student Surulivelan, Mathanraja team, Poojasri, Abinaya S team presented as shown in Figure 1.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for conversion of NFA to DFA.	The students will be able to identify the need for analysis of complex engineering problems using FA	Able to design the process flow using Finite Automaton	Able to analyze the data for Pumping lemma for Finite Automata	The students will be able to follow ethics for solving problems	The students able to function as a team for conversion of FA to RE and RE to FA.	The students will be able to communicate one another and can able to present	The students able to learn and implement the concept in Natural Language Processing	The students able to apply mathematical and computing skills.	The students can Exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- Images / Screenshot of the practice:



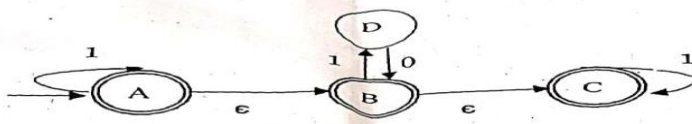
Figure 1: Collaboration Activity

1. Consider NFA $N=(Q,\Sigma,\delta,Q,F)$ where $Q=\{1,2,3\}$, $\Sigma=\{a,b\}$, $q=1$, $F=\{2\}$ and δ is given by the following table.

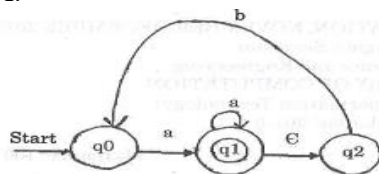
	a	b	c
1	3	$\varnothing$	2
2	$\{1\}$	$\varnothing$	$\varnothing$
3	$\{2\}$	$\{2,3\}$	$\varnothing$

Convert the NFA(N) into the DFA(M) that accepts the same language.

2. Convert the following ϵ -NFA to NFA and then convert the resultant NFA to DFA



3. Construct a epsilon NFA from the regular expression
(i) $(b/a)^*baa$
4. Construct ϵ -NFA for the following regular expression $(11+00)^*1(0+1)^*1$.
5. Convert the ϵ -NFA to DFA.



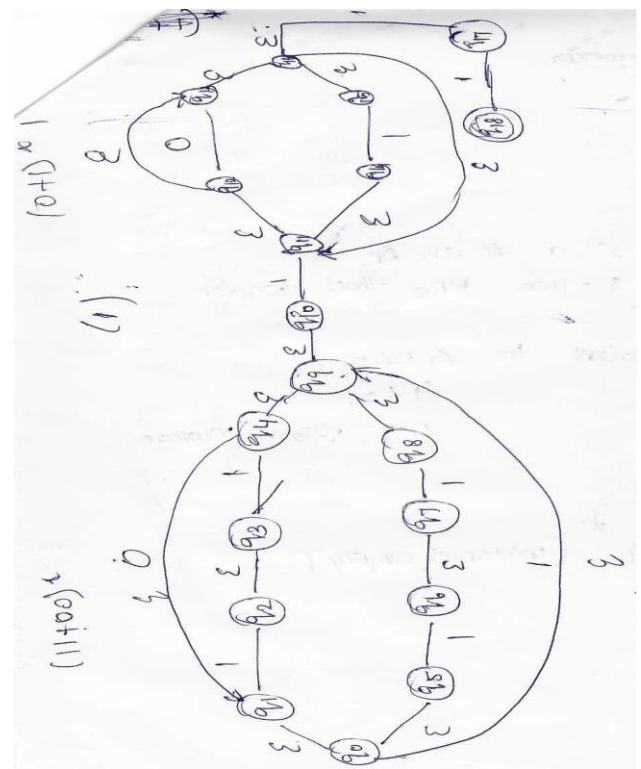
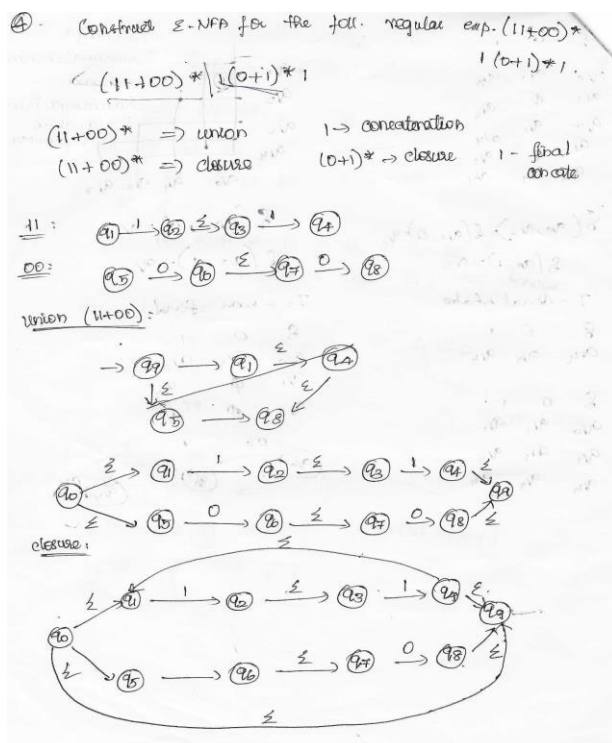
6. Determine DFA from the given NFA

$M = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_1\})$ where δ is given by
 $\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_1\}$, $\delta(q_1, 0) = \varnothing$, $\delta(q_1, 1) = \{q_0, q_1\}$

7. Construct DFA equivalent to NFA $\{p.q.r.s\}, \{0,1\}, \delta, p, \{s\}$ where δ is defined as

	0	1
$\rightarrow p$	$\{p.q\}$	$\{p\}$
q	$\{r\}$	$\{r\}$
r	$\{s\}$	-
s	$\{s\}$	$\{s\}$

Figure 2 Problems given to Students



Transition table

δ	0	1
q_0	$\{q_0, q_1\}$	q_1
q_0, q_1	$\{q_0, q_1\}$	$\{q_0, q_1\}$
q_1	ϕ	$\{q_0, q_1\}$

Diagram:-

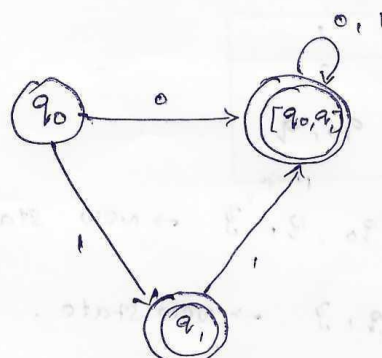


Figure 3 Solution given by the Student Mr. J.M. Thiraviyakumar and Team

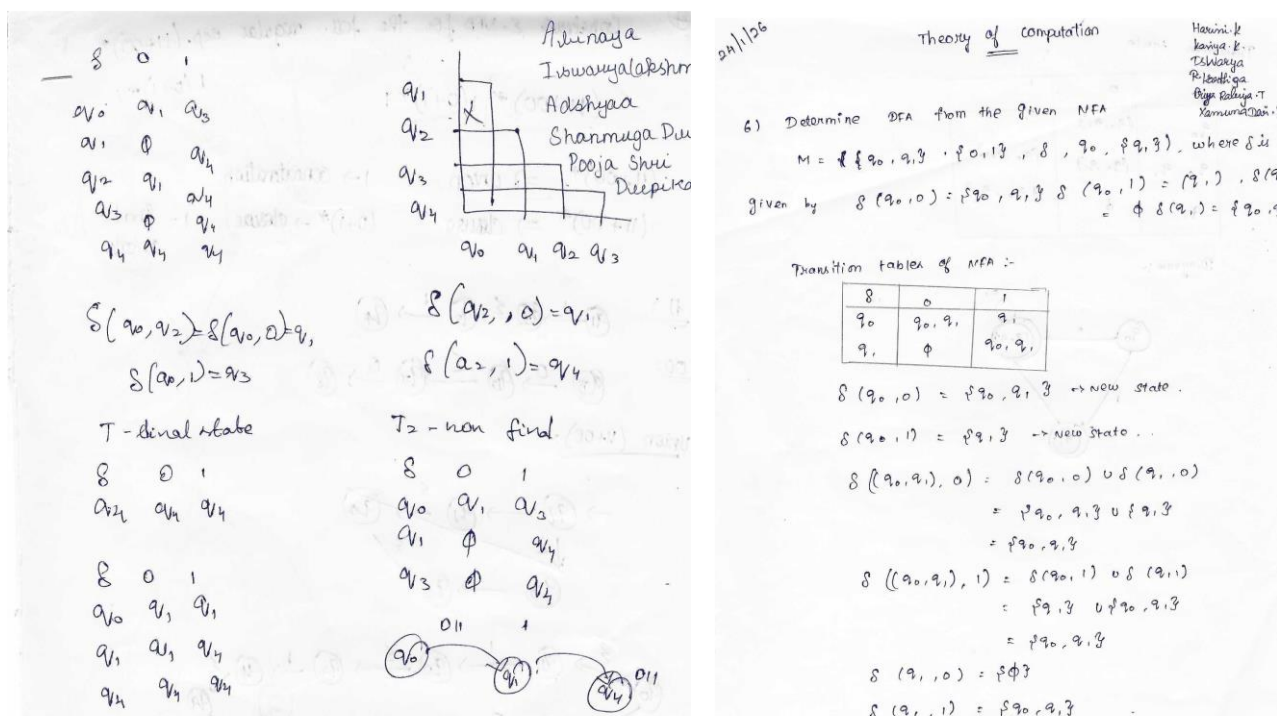


Figure 4 Solution given by the Students Ms. R. Abinaya and Ms.K. Harini Team
Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Students reported that the **collaborative activity enhanced their understanding** of automata conversions, especially subset construction in NFA to DFA.
- Hands-on practice with examples improved their **logical thinking and exam readiness**.
- The collaborative technique effectively promoted **student engagement and interaction**.
- Observers noted improved **student communication skills and analytical ability** during group problem solving.

❖ **Benefit of the practice:**

- Students collaboratively analyze automata diagrams, identify states, and perform conversions step-by-step, which strengthens logical thinking and algorithmic reasoning.
- Stronger students assist others, creating a supportive learning atmosphere and improving overall class performance.
- Working in groups develops collaboration, leadership, and accountability—important professional skills

❖ **Challenges faced in implementation:**

- Many students initially struggled to understand ϵ -closures, subset construction, and state mappings, which affected their confidence during group activities.
- Converting NFA to DFA is computationally intensive, and completing full examples within limited class time was challenging.

References:

- ❖ <https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning>
- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>

Signature of Faculty Member

HOD